Course Code: EEE 1261

Full Marks: 15

CT – 2

Course Title: Electronic Devices and Circuits

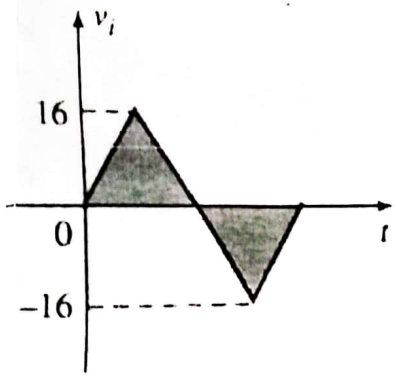
Time: 35 Minutes

Mark

Q.1 (a) Define a rectifier and explain the different types of rectifiers. 4

(b) Design a full-wave rectifier using a bridge rectifier and draw its input and output waveforms. 5

(c) Determine V_0 for the network of Fig. 1 (c), where V_1 and V_2 represent the last digits of your Student ID. 6



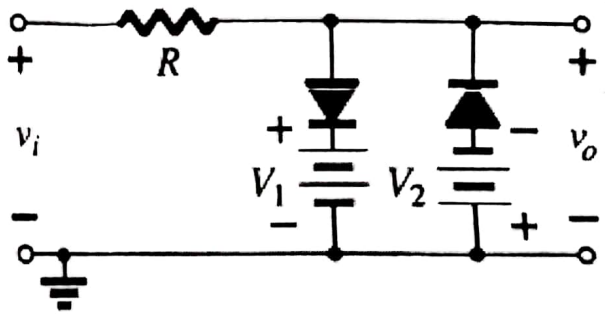
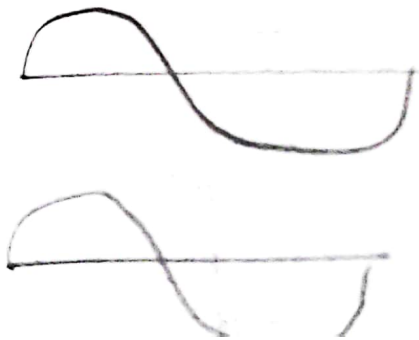


Fig. 1 (c)



CT-01

CT – 1

Course Code: EEE 1261 **Course Title: Electronic Devices and Circuits**

Full Marks: 20 **Time: 35 Minutes**

Mark

Q.1 (a) Explain the concepts of majority and minority charge carriers in semiconductors, specifying their roles in both n-type and p-type semiconductors. **5**

(b) Design a logic gate by using the diode that produces the waveform shown in Fig. 1 (b). **5**

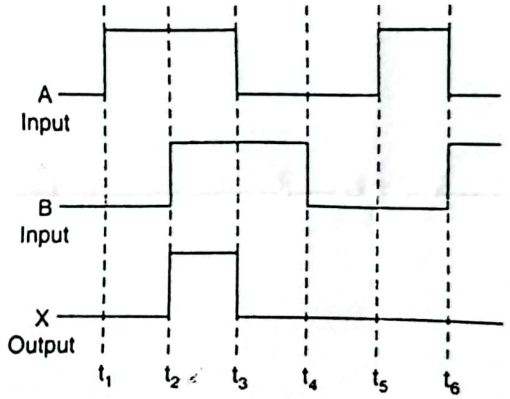


Fig. 1 (b).

(c) Determine the level of *Vo* for each network of Fig. 1 (c). **5**

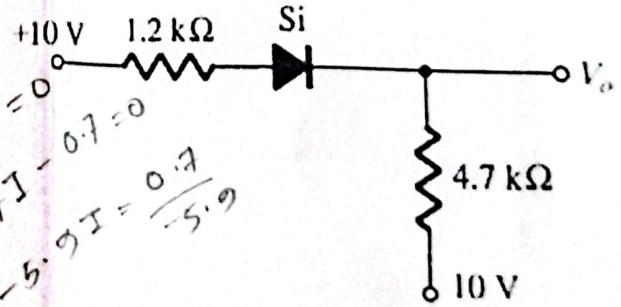


Fig. 1 (c).

$10 - IR_1 - 0.7 - IR_2 - 10 = 0$

$\Rightarrow -IR_1 - IR_2 - 0.7 = 0$

$\Rightarrow -I1.2 - 4.7I - 0.7 = 0$

$\Rightarrow -5.9I = \frac{0.7}{-5.9}$

$+10 + 4.7I$

$+10 + 4.7I + V_o$

CT-03

Course Code: EEE 1261 | CT – 3 | Course Title: Electronic Devices and Circuits

Full Marks: 15 | Time: 35 Minutes

			Mark
Q.1	(a)	Define the Zener diode and describe its characteristics.	3
	(b)	For the Zener diode network of Fig. 1(b), determine V_L, V_R, I_Z, and P_Z.	5

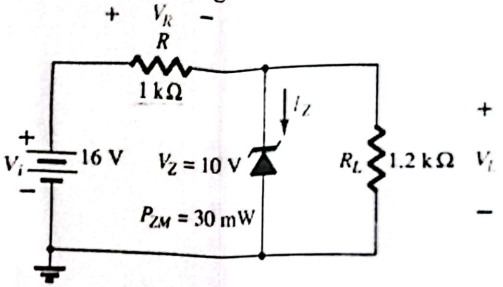


Fig. 1(b)

	(c)	Draw the output waveshape for the following circuit of Fig. 1 (c). Where, $V = 35V, C = 50\mu F, R = 10K\Omega$ and V_1 is the last digit of your Student ID.	7

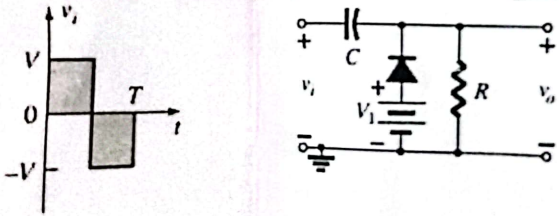
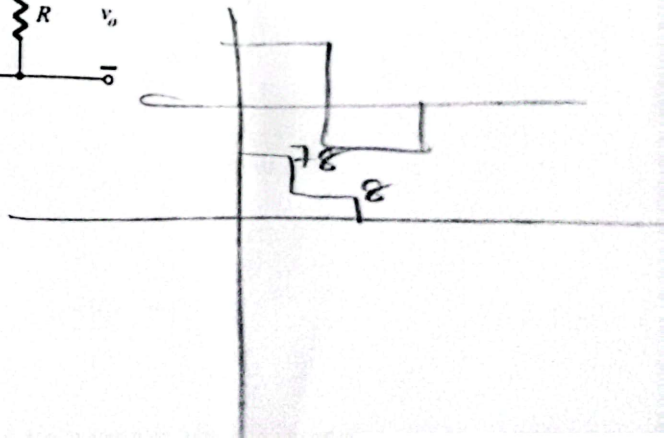


Fig. 1 (c)



CT-04

CT – 4

Course Code: EEE 1261 | Course Title: Electronic Devices and Circuits

Full Marks: 15 | Time: 35 Minutes

			Mark
Q.1	(a)	Define the following terms related to BJTs: (i) Base, (ii) Emitter, and (iii) Collector.	3
	(b)	Illustrate the circuit diagrams for the NPN BJT in common-base, common-emitter, and common-collector configurations, labeling all key components and terminals.	5
	(c)	For the fixed-bias configuration of Fig. 1 (c), determine: a) I_{B_Q} b) I_{C_Q} c) V_{CE_Q} d) V_C e) V_B f) V_E	7

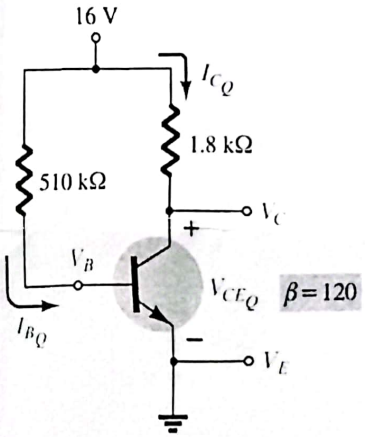
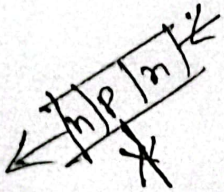
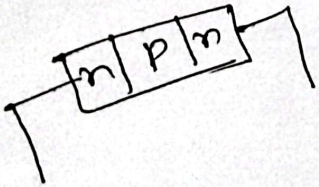


Fig. 1 (c)





baipotes

Here,

$R_B = 510\,k\Omega$

$V_{BE} = 0.7\,V$

$V_{CC} = 16\,V$

ignment: Assignment

Mark

5

Determine V_o and I_D for the networks of Fig. 1.

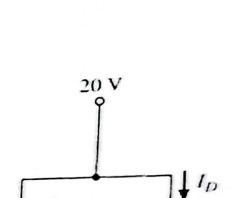


Fig. 1.